AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the CLOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.

II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.

III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

> **On verification.** The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 5	**Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.
STANDOUT SUCCESS	Case 18	**U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain 2017	T·K·N
2	Marine Corps Training in the INDOPACOM AOR ongoing	T·K
3	F-35 Sustainment & Maintainer Shortage ongoing	T·K·D
4	Military Fratricide — Desert Storm to Afghanistan 1991 – 2014	T·H·K
5	Operation Eagle Claw 1980	T·K
8	Patriot Missile / Dhahran 1991	D·H·K
17	GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data 2022 (GAO-22-104533)	G·K
18	U.S. Nuclear Navy / Rickover Training Model 1954 – present	T·K·N
21	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard 2020 revision (series since 1968)	G·K
33	GIFT and the Adoption Gap 2012 – present	K·G·N
37	Navy Surface Warfare Readiness Reform 2018 – present	T·K·N

42	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D
54	USS Vincennes & Iran Air Flight 655	1988	H·T
64	Stanislav Petrov / 1983 False Alert	1983	H·T
138	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
141	V-22 Osprey	1991 – present	T·H·N
167	9/11 Intelligence Sharing Failures	1996 – 2001	G·K

AVIATION

STANDOUT FAILURE — **Case 7**

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS — **Case 70**

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

6	AeroPerú Flight 603	1996	H·T
7	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
12	Boeing 737 Rudder Hardovers	1991, 1994	H·D
28	Air France Flight 447	2009	T·H
55	Kegworth / British Midland 92	1989	T·H·K
56	Asiana Airlines Flight 214	2013	T·H
57	Helios Airways Flight 522	2005	T·H
58	TransAsia Airways Flight 235	2015	T·H
63	Eastern Air Lines Flight 401	1972	H·T
68	Air Canada Chatbot Liability — Delegation Without Revocation	2022 – 2024	D·K·N
70	Crew Resource Management & CAST	1981 – present	T·H·N
72	Korean Air Safety Transformation	2000 – present	T·N
82	Colgan Air Flight 3407	2009	T
83	Atlas Air Flight 3591	2019	T·K
88	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N
111	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N

118	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
119	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson	1981 – 2008 (TCAS II Version 7.1)	H·K·G
164	NASA Saturn V Documentation	1972 – present	K
175	Singapore Airlines Safety Transformation	1980s – present	T·N

HEALTHCARE

STANDOUT FAILURE — **Case 61** — **EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS — **Case 109** — **WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

20	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G·K
32	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T·K·N
34	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K·G·N
59	Therac-25	1985 – 1987	H·D·G
61	EHR / CPOE Implementation	2005 – present	H·D·G
71	Keystone ICU / Pronovost Checklist	2004 – present	T·N
74	TREWS / Sepsis Watch	2018 – 2022	H·K·G
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff	2024 – 2025	H·K·N
79	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H·D
86	Mid Staffordshire NHS Foundation Trust	2005 – 2009	G·N·K
93	Epic Sepsis Model	2017 – 2021	D·G·N
102	Medical Errors as Systemic Failure	1999 – present	T·H·N·K·G
105	Vioxx Withdrawal	1999 – 2004	G·D
109	WHO Surgical Safety Checklist	2008 – present	T·N
112	Bristol Heart Babies Reform	1984 – present	G·N
125	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T·D·N
126	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T·D·N

EDUCATION AT SCALE

STANDOUT FAILURE

Case 149

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS

Case 110

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

ENERGY

STANDOUT FAILURE	Case 69	**Three Mile Island (1979).** Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.
STANDOUT SUCCESS	Case 172	**INPO and the Nuclear Academy (1979–).** The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

11	Sago Mine Disaster	2006	N·T·K
60	Northeast Blackout	2003	H·K
69	Three Mile Island	1979	T·H
85	Deepwater Horizon	2010	T·N·K
87	Upper Big Branch Mine Explosion	2010	N·G·K
137	Texas City BP Refinery Explosion	2005	N·T·K·G
142	Davis-Besse Reactor Head Corrosion	2002	N·K·G
143	Fukushima Daiichi	2011	N·G·K
154	Camp Fire / PG&E	2018	G·N·K
172	INPO and the Nuclear Academy	1979 – present	T·K·G

GOVERNMENT

STANDOUT FAILURE	Case 162	**Australia Robodebt (2016–20).** An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.
NOTE	Case 111	No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

14	Healthcare.gov Launch	2013	K·T·G
101	VA Wait-Time Scandal	2014	G·K·N
103	LIBOR Manipulation	2003 – 2012	G·N
131	Predictive Policing — PredPol	2011 – present	G·H·D

166	Bernard Madoff / SEC Failures	1992 – 2008	G·K·N

INDUSTRIAL

STANDOUT FAILURE — Case 147

Bhopal (1984). Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS — Case 73

Toyota Production System / Andon Cord (1950s–). Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

30	Kodak Digital Camera Stagnation	1975–2012	D·K·N
73	Toyota Production System / Andon Cord	1950s – present	N·G
84	Takata Airbag Inflators	2008 – 2023	D·G
92	Volkswagen Dieselgate	2015	D·G
139	GM Ignition Switch	2002 – 2014	D·G
147	Bhopal	1984	T·K·N·G
148	Grenfell Tower	2017	G·T·K·N
153	Hyatt Regency Walkway Collapse	1981	D·G

TECHNOLOGY

STANDOUT FAILURE — Case 66

UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.

NOTE — Case 132

The technology dataset's success story is filed under Healthcare: AlphaFold (Case 132), the contemporary worked example of computational capability-engineering done as a discipline.

10	Knight Capital Trading Loss	2012	D·K
31	BlackBerry Touchscreen Inertia	2007–2013	D·N·K
66	UK Post Office Horizon Scandal	1999 – 2015	G·H·K
80	AI-Augmented Coding Tools	2021 – present	T·H
91	Wells Fargo Fake Accounts	2011 – 2016	G·N

AI IN EDUCATION

K-12 EDUCATION

K-12 MATHEMATICS

K-12 SCIENCE

AIR TRAFFIC MANAGEMENT

ANESTHESIOLOGY

AUTONOMOUS SYSTEMS

STANDOUT FAILURE — Case 62

Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE — Case 194

No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 194 (**The Discipline We Build Next**) for the open question this volume closes on.

AUTONOMOUS VEHICLES

AVIATION INFRASTRUCTURE

AVIATION SAFETY

CLINICAL CARE

CLINICAL MEDICINE

CLINICAL/TRANSLATIONAL RESEARCH

CORPORATE L&D

CRIMINAL JUSTICE

DATA GOVERNANCE

DIGITAL IDENTITY

E-GOVERNMENT

ED-TECH

EDUCATION (NORWAY)

EDUCATION AT SCALE

FINANCE

FINANCIAL SERVICES

GENERAL AVIATION

GLOBAL HEALTH

GOVERNMENT

GOVERNMENT/WELFARE (NETHERLANDS)

HEALTH PROFESSIONS EDUCATION

HEALTHCARE/ONCOLOGY

HIGHER EDUCATION

HIGHER EDUCATION (AFRICA)

HIGHER EDUCATION (BRAZIL)

NAVAL ENGINEERING

NEUROSCIENCE

NEUROSCIENCE/CONNECTOMICS

NUCLEAR ENGINEERING

NUCLEAR POWER

NURSING EDUCATION

PHARMA SAFETY

RAIL TRANSPORT

SOFTWARE ENGINEERING EDUCATION

SOFTWARE SUSTAINMENT

SPACE / AEROSPACE

STANDOUT FAILURE	Case 140	**Challenger & Columbia (1986 / 2003).** Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 18	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 18) for the safety-engineering counterpart that aerospace did not build.

SURGERY

TECHNOLOGY

TUTORING

WORKFORCE DEV

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

34 cases

1	USS Fitzgerald & USS John S. McCain
7	Boeing 737 MAX / MCAS
14	Healthcare.gov Launch
16	EU Human Brain Project — Top-Down Vision That Unraveled
17	GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data
19	Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable
21	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard
22	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule
23	FAA NextGen and the ADS-B Out Transition
24	Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed
25	Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change
26	INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding
27	Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)
28	Air France Flight 447
33	GIFT and the Adoption Gap
34	Implementation Science in Healthcare — The 17-Year Gap
38	Cognitive Tutor / Carnegie Learning
39	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model
42	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks
69	Three Mile Island

LEN 2

53 cases

116 Anesthesia Monitoring Standards and the APSF — The Mortality Decline

120 Bar-Code Medication Administration — Cue at the Point of Care

121 Surgical Skill Video Peer-Rating Predicts Complications

122 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

127 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive

128 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

144 CrowdStrike Falcon Outage

185 Singapore SkillsFuture — National Workforce Capability at Scale

LEN 3

51 cases

1 USS Fitzgerald & USS John S. McCain

3 F-35 Sustainment & Maintainer Shortage

12 Boeing 737 Rudder Hardovers

18 U.S. Nuclear Navy / Rickover Training Model

26 INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

29 Amazon Hiring AI — Trained Bias, Deprecated 2018

30 Kodak Digital Camera Stagnation

31 BlackBerry Touchscreen Inertia

45 Spaced Education RCTs in Medical Training

49 ASSISTments — National Replication and Long-Term Follow-Through

50 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

51 Zhang/Scardamalia — Knowledge Building Across Three Cohorts

52 Chen et al. — Rural China AI Devices and the Equity-Direction Finding

65 NASA EVA-23 — Water Intrusion Inside a Spacesuit

70 Crew Resource Management & CAST

85 Deepwater Horizon

86 Mid Staffordshire NHS Foundation Trust

88 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap

98 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

99 COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

100 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

LEN 4

65 cases

32	ACGME 80-Hour Resident Duty-Hour Reform
35	Training Transfer — The Gap Between Learning and Doing
37	Navy Surface Warfare Readiness Reform
38	Cognitive Tutor / Carnegie Learning
43	I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer
46	High-Impact Learning System — Engineering the Environment, Not Just the Event
47	PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption
70	Crew Resource Management & CAST
71	Keystone ICU / Pronovost Checklist
74	TREWS / Sepsis Watch
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff
82	Colgan Air Flight 3407
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87	Upper Big Branch Mine Explosion
89	Tennessee Voluntary Pre-K Study
90	Algorithmic Bias in Educational Predictive Analytics
91	Wells Fargo Fake Accounts
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93	Epic Sepsis Model
94	Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity
95	Racial Bias in Pain Assessment — The False-Belief Mechanism
97	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism
101	VA Wait-Time Scandal
102	Medical Errors as Systemic Failure
103	LIBOR Manipulation
104	Atlanta Public Schools Cheating Scandal

183 Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact

184 CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver

185 Singapore SkillsFuture — National Workforce Capability at Scale

186 Australian Hospital-Pharmacy Technician Role Redesign

188 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

190 CARE Principles — Indigenous Data Governance as a Replaced Regime

193 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map

LEN 5

85 cases

1 USS Fitzgerald & USS John S. McCain

2 Marine Corps Training in the INDOPACOM AOR

3 F-35 Sustainment & Maintainer Shortage

4 Military Fratricide — Desert Storm to Afghanistan

5 Operation Eagle Claw

6 AeroPerú Flight 603

7 Boeing 737 MAX / MCAS

8 Patriot Missile / Dhahran

9 Mars Climate Orbiter — Unit Mismatch

10 Knight Capital Trading Loss

11 Sago Mine Disaster

12 Boeing 737 Rudder Hardovers

13 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training

14 Healthcare.gov Launch

15 Ariane 5 Flight 501

17 GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data

18 U.S. Nuclear Navy / Rickover Training Model

19 Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

20 California Nurse Staffing Ratios — Legislating a Capability Requirement

21 MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

27 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

28 Air France Flight 447

LEN 6

17 cases

LEN 7

106 cases

179 Implementation-Science Training — Stated Goals Outrunning Operational Practice

180 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

181 LALA — Building Learning-Analytics Governance Capacity Across Latin America

182 Norway's National Expert Commission on Learning Analytics

188 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

189 Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

192 Cruise Robotaxi — Pedestrian Drag

193 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map

LEN 8

92 cases

1 USS Fitzgerald & USS John S. McCain

2 Marine Corps Training in the INDOPACOM AOR

3 F-35 Sustainment & Maintainer Shortage

4 Military Fratricide — Desert Storm to Afghanistan

5 Operation Eagle Claw

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29 Amazon Hiring AI — Trained Bias, Deprecated 2018

32 ACGME 80-Hour Resident Duty-Hour Reform

33 GIFT and the Adoption Gap

34 Implementation Science in Healthcare — The 17-Year Gap

36 Growth-Mindset National Experiment — Heterogeneous Effects

40 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

LEN 9

28 cases

LEN 10

19 cases

APPENDIX · REFERENCES BY CASE

References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in `casebook/verification-log.md`. Future editions will fill in remaining locators as the per-case review closes them out.

> **Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

CASE 1 USS Fitzgerald & USS John S. McCain 2017

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3. NTSB, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), DCA17PM018. — Retrieved from: ____________________
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CASE 2 Marine Corps Training in the INDOPACOM AOR

ongoing

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CASE 3 F-35 Sustainment & Maintainer Shortage

ongoing

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3. GAO-23-105341 (2023) — sustainment shortfall traced to maintainer shortages, lack of military access to technical data, and contractor dependency. — Retrieved from: ____________________

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CASE 4 Military Fratricide — Desert Storm to Afghanistan

1991 – 2014

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